Deep Learning Notes

Session 1 — Taught by Kanhaiya, Corrected & Expanded

1. Neuron

Your definition: A neuron is a bridge for human and computer understanding.

Corrected definition:

A neuron is a mathematical unit that takes inputs, multiplies each by a weight, adds a bias, and passes the result through an activation function.

Formula:

```
output = activation(weight * input + bias)
```

> **Key insight:** This is inspired by biological neurons in the brain, but in math terms, it is just a weighted sum passed through a function.

2. Activation Functions

Your intuition: Activation helps models learn curves and non-linear boundaries. Without it, layers stay linear.

Correct — the technical term is: Non-linearity. Remember this word.

Activations you covered:

• **Sigmoid** — Output between 0 and 1. Used in binary classification. Problem: Vanishing gradient.

• **ReLU** (Rectified Linear Unit) — Filters negative values to 0. Problem: Dead neurons (neurons stuck at 0 forever).

• **Softmax** — You forgot the name. Converts outputs into probabilities that sum to 1. Used for multi-class classification.

What you missed:

• **Tanh** — Output between -1 and 1. Better than sigmoid in some cases.

• **Leaky ReLU** — Fixes the dead neuron problem of ReLU by allowing small negative values.

3. Forward Pass

Your definition: Input features pass through hidden layers, then activation, then output.

Correction: Activation is not separate from layers — it is inside each layer.

Correct flow:

```
input → [weights + bias + activation] → [weights + bias + activation] →
output
```

Each hidden layer applies weights, bias, AND activation together.

4. Loss Function

Your definition: Log loss blames weights and bias for error.

Correction: Blaming weights is backpropagation's job, not the loss function's job.

Loss function's actual job:

Measure how wrong the prediction is and give a single number representing the error.

- **MSE** (Mean Squared Error) — Used for regression.
- **Log Loss / Binary Cross Entropy** — Used for classification.

Important: Log Loss is also called Binary Cross Entropy. Both names refer to the same thing.

5. Backpropagation

Your definition: Backpropagation assigns responsibility for error and updates weights and bias.

Good intuition — but the mechanism was missing.

Full explanation:

Backpropagation uses calculus (chain rule) to calculate how much each weight contributed to the error.

3 things people confuse — keep them separate:

- **Backpropagation** — Calculates gradients (how much each weight is responsible).
- **Gradient Descent** — Uses those gradients to update weights.
- **Learning Rate** — Controls how big each update step is.

Homework Question for Next Session

Why does sigmoid cause the vanishing gradient problem?

Think about it. Come and teach me the answer tomorrow.